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Interim Considerations: Preparing for the Potential Management of Anaphylaxis after COVID-19 Vaccination

Recognizing and Responding to Anaphylaxis  [1 page, 508]

Summary of recent changes (last updated March 22, 2024)

- Updated recommended supplies for vaccination settings
- Updated guidance for observation periods following COVID-19 vaccination
- Included option of 0.1 mg epinephrine auto-injector for patients weighing 7.5 kg to <15kg
- Updated patient counseling for patients with allergy-related contraindications and precautions
- Updated VAERS reporting recommendations
- Updated links to FDA vaccine pages
- Removed information related to Janssen

Key Points

- Anaphylaxis, an acute and potentially life-threatening allergic reaction, has been reported rarely following COVID-19 vaccination.
- Under the [Food and Drug Administration \(FDA\) approvals and Emergency Use Authorizations](#)  for COVID-19 vaccines, appropriate medical treatment for severe allergic reactions must be immediately available at any site administering the vaccine in the event that an acute anaphylactic reaction occurs following administration of a COVID-19 vaccine.
- These interim considerations provide information on preparing for the initial assessment and potential management of anaphylaxis following COVID-19 vaccination.

Overview

Anaphylaxis, an acute and potentially life-threatening allergic reaction, has been reported rarely following COVID-19 vaccination. These interim considerations provide recommendations on assessment and potential management of anaphylaxis following COVID-19 vaccination. Detailed information on CDC recommendations for vaccination, including contraindications and precautions to vaccination, can be found in the Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States. Patients should be screened prior to receipt of each vaccine dose, and those with a contraindication should not be vaccinated.

Personnel, medications, and supplies for assessing and managing anaphylaxis

Healthcare personnel who are trained and qualified to recognize the signs and symptoms of anaphylaxis, as well as administer intramuscular epinephrine, should be available at the vaccination location at all times. Vaccination locations that anticipate vaccinating people with COVID-19 vaccines, including mass vaccination clinics, should plan adequate staffing and supplies (including epinephrine) for the assessment and potential management of anaphylaxis.

The following emergency equipment should be immediately available for the assessment and management of anaphylaxis.

Should be available at all locations	If feasible, include at locations (not required)
Epinephrine (e.g., prefilled syringe, autoinjector)*	Pulse oximeter
Blood pressure monitor†	Oxygen
Timing device to assess pulse	Bronchodilator (e.g., albuterol)
	H1 antihistamine (e.g., diphenhydramine, cetirizine)†
	H2 antihistamine (e.g., famotidine, cimetidine)
	Intravenous fluids
	Intubation kit
	Pocket mask with one-way valve (also known as cardiopulmonary resuscitation [CPR] mask) sized for adults and children

*COVID-19 vaccination locations should have **at least 3 doses** of age-appropriate epinephrine available at all times, and the ability to quickly obtain additional doses to replace supplies after epinephrine is administered to a patient. Locations that are administering COVID-19 vaccines to children <12 years should have age-appropriate supplies, including age-appropriate epinephrine dosing. People with a history of anaphylaxis who carry an epinephrine autoinjector could be encouraged to bring it to their vaccination appointment. Detailed information on storage, handling, administration, and dosage considerations is available in the package inserts for [epinephrine](#) (e.g., [EpiPen](#)® or [AUVI-Q](#)®). Expired epinephrine or epinephrine that appears to be in unacceptable condition (per the manufacturer's package inserts) should be replaced.

†Antihistamines may be given as adjunctive treatment but should not be used as initial or sole treatment for anaphylaxis. Additionally, caution should be used if oral medications are administered to people with impending airway obstruction.

*Either an automated or a manual blood pressure monitor, with appropriate cuff sizes, is acceptable. If a manual blood pressure monitor is used, a stethoscope should also be available.

Post-vaccination observation periods

Providers should consider observing all vaccine recipients for 15 minutes after vaccination for syncope, per the Advisory Committee on Immunization Practices' General Best Practice Guidelines.

Additionally, to monitor for allergic reactions, providers should consider observing people with the following precautions to a previously administered COVID-19 vaccine type for 30 minutes if a subsequent dose of the same vaccine type is administered:

- History of a non-severe, immediate (onset less than 4 hours) allergic reaction after administration of a previous dose of one COVID-19 vaccine type*.
- History of a diagnosed non-severe allergy to a component of the COVID-19 vaccine

See Contraindications and precautions for more information.

* The mRNA COVID-19 vaccines (Moderna and Pfizer-BioNTech) are one type of COVID-19 vaccine, and the protein subunit vaccine (Novavax) is another type of COVID-19 vaccine.

Early recognition of anaphylaxis

Because anaphylaxis requires immediate treatment, diagnosis is primarily made based on recognition of clinical signs and symptoms. Signs and symptoms in adults and children include:

- **Respiratory:** sensation of throat closing or tightness, stridor (high-pitched sound while breathing), hoarseness, respiratory distress (such as shortness of breath or wheezing), coughing, trouble swallowing/drooling, nasal congestion, rhinorrhea, sneezing
- **Gastrointestinal:** nausea, vomiting, diarrhea, abdominal pain, or cramps
- **Cardiovascular:** dizziness; fainting; tachycardia (abnormally fast heart rate); hypotension (abnormally low blood pressure); pulse difficult to find or "weak"; cyanosis (bluish discoloration); pallor; flushing
- **Skin/mucosal:** generalized hives; widespread redness; itching; conjunctivitis; or swelling of eyes, lips, tongue, mouth, face, or extremities
- **Neurologic:** agitation; convulsions; acute change in mental status; sense of impending doom (a feeling that something bad is about to happen)
- **Other:** sudden increase in secretions (from eyes, nose, or mouth); urinary incontinence



Anaphylaxis should be considered when signs or symptoms are generalized (i.e., if there are generalized hives or more than one body system is involved) or are serious or life-threatening in nature, even if they involve a single body system (e.g., hypotension, respiratory distress, or significant swelling of the tongue or lips).

Symptoms of anaphylaxis often occur within 15-30 minutes of vaccination, though it can sometimes take several hours for symptoms to appear. Early signs of anaphylaxis can resemble a mild allergic reaction, and it is often difficult to predict whether initial, mild symptoms will progress to become an anaphylactic reaction. In addition, symptoms of anaphylaxis might be more difficult to recognize in infants and toddlers, people with communication difficulties, such as long-term care facility residents with cognitive impairment, those with neurologic disease, or those taking medications that can cause sedation. Not all symptoms listed above are necessarily present during anaphylaxis, and not all patients have skin reactions.

More information on potential characteristics of allergic reactions, vasovagal reactions, and vaccine side effects following COVID-19 vaccination can be found in the Appendix.

If anaphylaxis is suspected, administer epinephrine as soon as possible, contact emergency medical services, and transfer patients to a higher level of medical care. In addition, instruct patients to seek immediate medical care if they develop signs or symptoms of an allergic reaction after their observation period ends and they have left the vaccination location.

Management of anaphylaxis at a COVID-19 vaccination location

If anaphylaxis is suspected, take the following steps:

- Rapidly assess airway, breathing, circulation, and level of consciousness (mental activity).
- Call for emergency medical services (EMS).
- Ensure the patient is in a stable position, e.g. seated or laying.
- Epinephrine (1 mg/ml aqueous solution [1:1000 dilution]) is the first-line treatment for anaphylaxis and should be administered immediately, as an intramuscular injection.
 - A dose of epinephrine may be repeated approximately every 5-15 minutes if symptoms do not improve or if they return while waiting for EMS. The number and timing of epinephrine doses should be recorded and communicated to EMS.
 - Because of the acute, life-threatening nature of anaphylaxis, there are no contraindications to epinephrine administration.

Weight-based intramuscular epinephrine dosing and administration for anaphylaxis

Weight	Preferred Method	Available Autoinjectors EpiPen and EpiPen Jr   AUVI-Q 	Weight-based Maximum Dose (Epinephrine (1 mg/ml aqueous solution [1:1000 dilution]) at dose of 0.01 mg/kg)*
>30 kg	Autoinjector, premeasured or prefilled syringe	• EpiPen 0.3 mg • AUVI-Q 0.3 mg	0.5 mg
15 kg to <30 kg	Autoinjector	• EpiPen Jr 0.15 mg • AUVI-Q 0.15 mg	0.3 mg
<15 kg	Weight-based dosing	• AUVI-Q 0.1 mg <i>If AUVI-Q 0.1 mg Autoinjector is unavailable</i> • EpiPen Jr 0.15 mg • AUVI-Q 0.15 mg	0.15 mg

*Weight-based dosing may only be possible in certain clinics or medical offices

Source: Campbell RL, Kelso JM. Anaphylaxis: Emergency treatment. In: UpToDate, Post TW (Ed), UpToDate, Waltham, MA.

- At the earliest opportunity, place the patient in a supine position (face up), with feet elevated, unless respiratory distress or upper airway obstruction is present, or the patient is vomiting. Do not delay administration of epinephrine.
- Antihistamines (e.g., H1 or H2 antihistamines) and bronchodilators do not treat upper airway obstruction (laryngeal edema) or hypotension and, thus, are not first-line treatments for anaphylaxis. Although they can help provide relief for hives and itching (antihistamines) or symptoms of respiratory distress (bronchodilators), in a patient with

anaphylaxis they should only be administered *after* epinephrine. Administration of antihistamines to COVID-19 vaccine recipients prior to vaccination to prevent anaphylaxis is not recommended.

- Because anaphylaxis may recur after patients begin to recover, **monitoring in a medical facility for several hours** is advised, even after complete resolution of symptoms and signs.

Considerations for anaphylaxis management in special populations

Children <12 Years

Facilities providing COVID-19 vaccines for children aged <12 years should ensure they have age and size appropriate emergency supplies on hand. Many symptoms of anaphylaxis in children are similar to those seen in adults. Infants and toddlers may also present with behavioral signs such as pulling or scratching at ears, inconsolable crying, or irritability, and need to be monitored closely for development of other symptoms of anaphylaxis.

Children <30 kg (66 lbs.) require lower dosing of epinephrine for management of anaphylaxis. Pediatric autoinjectors are available in 0.1 mg, 0.15 mg, and 0.30 mg doses. Epinephrine dosing in children is based on weight and not dependent on age, more details can be found in the previous section.

Older adults, including long-term care facility residents

There are no contraindications to the administration of epinephrine for the treatment of anaphylaxis. Although adverse cardiac events, such as myocardial infarction or acute coronary syndrome, have been reported in some patients who received epinephrine for treatment of anaphylaxis (particularly among older adults with hypertension and/or atherosclerotic heart disease), epinephrine is the first-line treatment for anaphylaxis. It is important that locations providing vaccination to older adults, including long-term care facility residents, have staff members available who are able to recognize the signs and symptoms of anaphylaxis. This will help not only to ensure appropriate and prompt treatment for patients with anaphylaxis, but also to avoid unnecessary epinephrine administration to patients who do not have anaphylaxis.

Pregnant women

Pregnant women with anaphylaxis should be managed in the same manner as non-pregnant women. As with all patients with anaphylaxis, they should be transported to a medical facility where they and their fetus can be closely monitored to ensure adequate perfusion.

Patient counseling

People with a contraindication to one COVID-19 vaccine type (Table 3) may receive the alternative COVID-19 vaccine type* in the usual vaccination setting. Consultation with an allergist-immunologist is encouraged to provide expert evaluation of the original allergic reaction, and depending on the outcome of the evaluation, reassess if administration of additional doses of the same vaccine type may be possible.

People with an allergy-related precaution to one COVID-19 vaccine type (Table 3) may receive the alternative COVID-19 vaccine type* in the usual vaccination setting. Vaccination with the same COVID-19 vaccine type may be considered on an individual basis; the same vaccine type should be administered in an appropriate setting and under the supervision of a health care provider experienced in the management of severe allergic reactions. An observation period of 30 minutes post-vaccination should be considered. Referral to an allergist-immunologist should be considered.

*The mRNA COVID-19 vaccines (Moderna and Pfizer-BioNTech) are one type of COVID-19 vaccine, and the protein subunit vaccine (Novavax) is another type of COVID-19 vaccine.

Reporting anaphylaxis

Anyone can report any adverse events, including anaphylaxis, that occur in a recipient following COVID-19 vaccination, to the [Vaccine Adverse Event Reporting System](#) (VAERS). Reporting is encouraged for any clinically significant adverse event, even if it is uncertain whether the vaccine caused the event. For detailed reporting requirements for VAERS, please see [Clinical Guidance for COVID-19 Vaccination | CDC](#). Refer to the [VAERS website](#) or call 1-800-822-7967 for more information on how to submit a report to VAERS.

Additional resources

[Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States](#)

[ACIP Vaccine Recommendations and Guidelines](#)

[Immunization Action Coalition: Medical Management of Vaccine Reactions in Adults](#)

[Immunization Action Coalition: Medical Management of Vaccine Reactions in Children and Teens](#)

[Moderna COVID-19 Vaccine | FDA](#)

[SPIKEVAX | FDA](#)

[Pfizer-BioNTech COVID-19 Vaccine | FDA](#)

[COMIRNATY | FDA](#)

[Novavax COVID-19 Vaccine, Adjuvanted | FDA](#)

References

[Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States](#)

Golden DBK, et al. "Anaphylaxis: A 2023 practice parameter update." *Ann Allergy Asthma Immunol*. 2023 Dec 5:S1081-1206(23)01304-2. doi: 10.1016/j.anai.2023.09.015. Epub ahead of print. PMID: 38108678.

Lieberman P, et al. "Anaphylaxis: A practice parameter update." *Annals of Allergy, Asthma & Immunology* 2015; 115(5): 341-384. doi: 10.1016/j.anai.2015.07.019.

Shaker MS, et al. "Anaphylaxis—a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis." *Journal of Allergy and Clinical Immunology* 2020;145(4):1082-1123. doi: 10.1016/j.jaci.2020.01.017.

Sicherer SH, Simons FER; SECTION ON ALLERGY AND IMMUNOLOGY. Epinephrine for First-aid Management of Anaphylaxis. *Pediatrics*. 2017;139(3):e20164006. doi:10.1542/peds.2016-4006

Appendix: Potential characteristics of allergic reactions, vasovagal reactions, and vaccine side effects following COVID-19 vaccination

In patients who experience post-vaccination symptoms, determining the etiology (including allergic reaction, vasovagal reaction, or vaccine side effects) is important to determine whether a person can receive further doses of the vaccine. The following table of signs and symptoms is meant to serve as a resource but might not be exhaustive, and patients might not have all signs or symptoms. Vaccination providers should use their clinical judgement when assessing patients to determine the diagnosis and management.

Characteristic	Allergic reactions (including anaphylaxis)	Vasovagal reaction	Vaccine side effects (local and systemic)
Timing after vaccination	Most occur within 15-30 minutes of vaccination	Most occur within 15 minutes	Median of 1 to 3 days after vaccination (with most occurring the day after vaccination)
Signs and symptoms			
Constitutional	Feeling of impending doom	Feeling warm or cold	Fever, chills, fatigue
Cutaneous	Skin symptoms present in ~90% of people with anaphylaxis, including pruritus, urticaria, flushing, angioedema	Pallor, diaphoresis, clammy skin, sensation of facial warmth	Pain, erythema, or swelling at injection site; lymphadenopathy in same arm as vaccination
Neurologic	Confusion, disorientation, dizziness, lightheadedness, weakness, loss of consciousness	Dizziness, lightheadedness, syncope (often after prodromal symptoms for a few seconds or minutes), weakness, changes in vision (such as spots of flickering lights, tunnel vision), changes in hearing	Headache
Respiratory	Shortness of breath, wheezing, bronchospasm, stridor, hypoxia	Variable; if accompanied by anxiety, might have an elevated respiratory rate	N/A
Cardiovascular	Hypotension, tachycardia	Variable; might have hypotension or bradycardia during syncopal event	N/A
Gastrointestinal	Nausea, vomiting, abdominal cramps, diarrhea	Nausea, vomiting	Vomiting or diarrhea might occur
Musculoskeletal	N/A	N/A	Myalgia, arthralgia
Vaccine and clinical management recommendations			
Can receive a subsequent dose of COVID-19 vaccine?	See Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States	Yes	Yes

Note: Severe allergic reactions include:

- Possible anaphylaxis, a progressive life-threatening reaction that typically includes urticaria but also with other symptoms such as wheezing, difficulty breathing, or low blood pressure
- Any angioedema affecting the airway (i.e., tongue, uvula, or larynx)
- Diffuse rash which also involves mucosal surfaces (e.g., Stevens-Johnson Syndrome)

Non-severe allergic reactions may include:

- Urticaria (hives) beyond the injection site
- Angioedema (visible swelling) involving lips, facial skin, or skin in other locations. NOTE: Any angioedema affecting the airway (i.e., tongue, uvula, or larynx) would NOT be in this category and is considered a severe allergic reaction

Previous Updates

Revisions made March 22, 2024

- Updated recommended supplies for vaccination settings
- Updated guidance for observation periods following COVID-19 vaccination
- Included option of 0.1 mg epinephrine auto-injector for patients weighing 7.5 kg to <15kg
- Updated patient counseling for patients with allergy-related contraindications and precautions
- Updated VAERS reporting recommendations
- Updated links to FDA vaccine pages
- Janssen vaccine no longer in use in the United States. Archived information from FDA is located at: [Janssen COVID-19 Vaccine EUA Fact Sheet for Healthcare Providers \(fda.gov\)](#) 

Revisions made Feb 11, 2022

- Added table for potential characteristics of allergic reactions, vasovagal reactions, and vaccine side effects following COVID-19 vaccination.

Revisions made Nov 3, 2021

- Considerations broadened to include children aged 5–11 years.

Revisions made March 3, 2021

- Considerations broadened to include use of Janssen (Johnson & Johnson) COVID-19 Vaccine.

Revisions made February 10, 2021

- Personnel, medications, and supplies for assessing and managing anaphylaxis: This section has been expanded to indicate that trained personnel qualified to recognize and treat symptoms of anaphylaxis should be available at vaccination locations at all times. The recommendations for medications and supplies have also been updated.
- Early recognition of anaphylaxis: This section has been updated to provide additional information related to anaphylaxis symptoms.
- Special populations: This section has been updated with considerations for anaphylaxis management of homebound people requiring home vaccination services.

Last Reviewed: April 2, 2024